NETFLIX

NETFLIX CUSTOMER CHURN PREDICTION

Machine Learning Report

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Netflix Customer Churn Prediction

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1. Problem Statement

The aim of this project is to predict whether a Netflix customer is likely to churn or continue their subscription.

Customer churn means that a customer stops using or cancels a service. Predicting churn can help a company identify customers who may leave and take steps to retain them.

In this project, I used customer details, subscription information and viewing behaviour to train and compare different machine learning models.

2. Dataset and Columns Used for Modelling

The dataset contains 5,000 customer records and 14 original columns. Calculated fields created for exploratory data analysis are excluded.

The original columns are:

1. customer_id
2. age
3. gender
4. subscription_type
5. watch_hours
6. last_login_days
7. region
8. device
9. monthly_fee
10. churned
11. payment_method
12. number_of_profiles
13. avg_watch_time_per_day
14. favorite_genre

The target column is churned: 0 means the customer stayed, and 1 means the customer churned. Customer ID is an identifier, not a prediction feature. Categorical columns were converted into numerical form.

3. Best Feature Selection

I used SelectKBest with the Chi-Square test to select the 7 most relevant features for predicting churn.

1. watch_hours
2. last_login_days
3. monthly_fee
4. subscription_type_Premium
5. number_of_profiles
6. avg_watch_time_per_day
7. payment_method_Crypto

These features include customer viewing behaviour, inactivity, subscription details and payment information.

The data was split into 70% training and 30% testing. StandardScaler was used to scale the selected features.

4. Machine Learning Models and Ranking

I trained and compared 10 classification algorithms:

1. Logistic Regression
2. Support Vector Machine – Linear
3. Support Vector Machine – Non-Linear
4. K-Nearest Neighbors (KNN)
5. Naive Bayes
6. Decision Tree
7. Random Forest
8. AdaBoost
9. CatBoost
10. XGBoost

The models were compared using accuracy scores. The ranking from the recorded comparison is shown on the next page.

Model Performance and Ranking

Model	Accuracy	Rank
XGBoost	97.40%	1
Random Forest	96.80%	2
AdaBoost	96.40%	3
Decision Tree	95.73%	4
KNN	92.27%	5
CatBoost	91.47%	6
SVM – Non-Linear	90.80%	7
SVM – Linear	89.20%	8
Logistic Regression	89.00%	9
Naive Bayes	82.27%	10

5. Final Model Selection

I selected XGBoost with 7 selected features as the final model. It achieved an accuracy score of 96.40%.

The final model predicts whether a customer is likely or unlikely to churn based on the information entered.